Light & Distance

Group Packet

Astrophysics Club 5th Meeting

Table of Constants

Constants and Conversion Factors	
Speed of light	$c = 3.00 \times 10^8$ m/s $= 3.00 \times 10^5$ km/s
Astronomical unit (Earth to Sun)	1 AU $= 1.496 \times 10^{11}$ m
Light-year	1 ly $= 9.46 \times 10^{15}$ m
One year	1 yr $= 3.16 \times 10^7$ s
Parsec	1 pc = 206,265 AU $= 3.09 \times 10^{16}$ m = 3.26 ly
Diameter of the Sun	$D_\odot = 1.39 \times 10^9$ m (radius 6.96×10^8 m)
Earth to Moon (average)	3.84×10^8 m = 384,400 km

Angles	
Degrees and radians	$180^\circ = \pi$ rad
Arcminutes and arcseconds	$1^\circ = 60'$, $1' = 60''$, $1^\circ = 3600''$
One radian	1 rad $= 57.3^\circ = 3438' = 206{,}265''$

Equations	
Distance traveled by light in time t	$d = ct$
Echo (there and back in time t)	$d = ct/2$
Right triangle, legs x, y, hypotenuse h	$h^2 = x^2 + y^2$
Arc length on a circle of radius r	$a = r\theta$ (θ in radians)
Small angle: object of size D at distance d	$\theta \approx D/d$ (θ in radians)
Parallax angle p (page 3)	$p \approx$ _____ (p in radians)
Distance in parsecs	d [pc] = _____ (p in arcseconds)

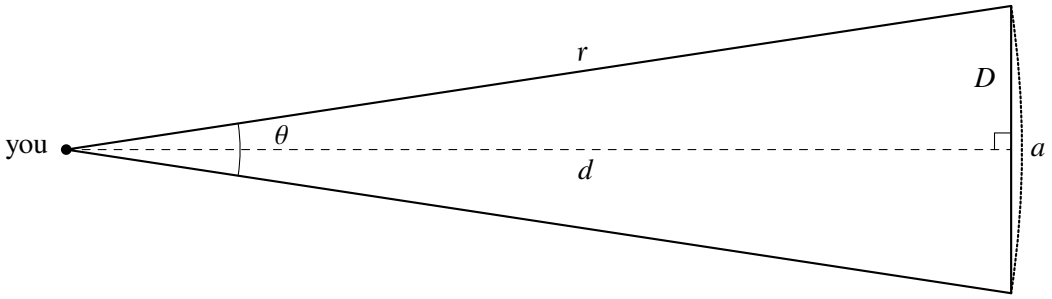
The Small-Angle Approximation

You look at an object of size D at distance d. It covers an angle θ at your eye. Measured in radians, the arc of radius r that θ cuts out has length

$$a = r\theta.$$

When θ is small, the arc a and the straight side D are almost the same length, and r is almost the same as d. So $D \approx a = r\theta \approx d\theta$, which gives

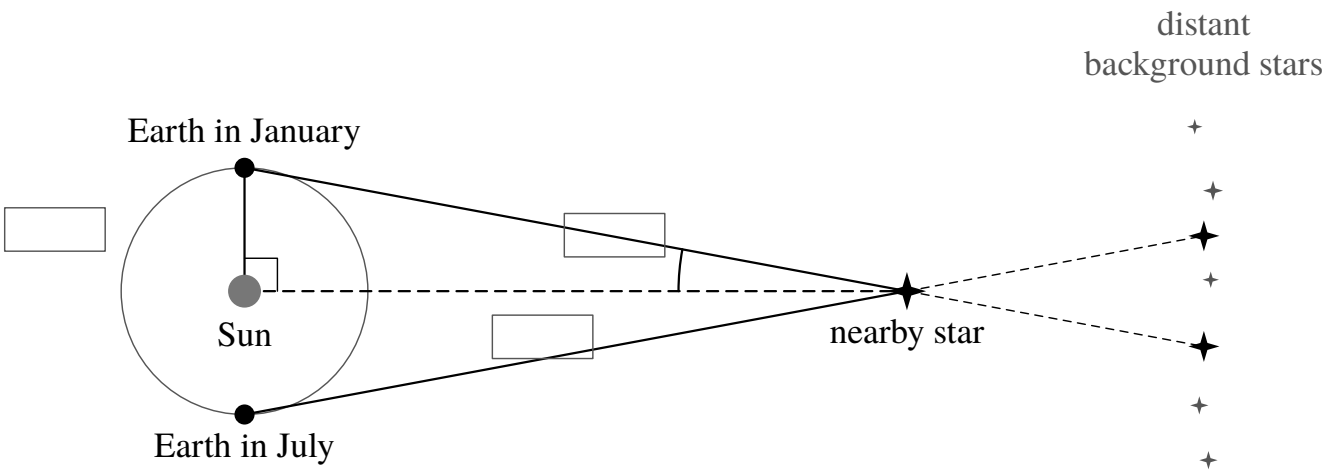
$$\theta \approx \frac{D}{d} \qquad (\theta \text{ in radians}).$$



The angle is drawn bigger than real. For a real object in the sky, θ is far smaller, so the dotted arc a and the straight side D are even closer together, and r and d are even closer in length.

Discover Parallax

Six months apart, Earth sees a nearby star from opposite sides of the Sun. The parallax angle p is the angle at the star between the Sun and Earth. Over six months the star seems to shift back and forth by a total angle of $2p$. The diagram is not to scale.



1. Fill in the three boxes on the diagram: the baseline from the Sun to Earth, the distance d from the Sun to the star, and the parallax angle p.

2. Look at the right triangle formed by the Sun, Earth in January, and the star. Write $\tan p$ in terms of 1 AU and d.

$$\tan p = _____$$

3. The parallax angle p is tiny. Use the small-angle approximation to find p in terms of 1 AU and d.

$$p \approx _____ \qquad (p \text{ in radians})$$

4. Convert to arcseconds with 1 rad = 206,265″ and 1 pc = 206,265 AU. Show that the distance in parsecs is

$$d\,[\text{pc}] = _____ \qquad (p \text{ in arcseconds})$$

Figures after A. Chen, "Telescopes," USAAAO Guide (2023), and B. W. Carroll and D. A. Ostlie, *An Introduction to Modern Astrophysics*, 2nd ed. (2007), Fig. 3.2.

Scratch Work